Jevi Waugh

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Experience

The University of Queensland — **School of Electrical Engineering and Computer Science**
Software Engineering Demonstrator — *July - current 2025*
Delivered content for CSSE1001, guide students in programming fundamentals, debugging, object-oriented design, and software engineering practices. Provide feedback to improve coding skills and confidence.

Apex Fincap
AI Integration Engineer — *Nov 2024 - Feb 2025*
I built AI systems that summarised financial data and auto-completed reports, improving efficiency and accuracy for brokers.

Queensland AI Safety Initiative (QuASI)
Technical Fellowship — *Feb - July 2025*
Completed an 8-week technical fellowship on AI safety, exploring Mechanistic Interpretability, Reinforcement Learning, Goal Misgeneralisation, and broader Alignment challenges.

Education

The University of Queensland — **St Lucia, Brisbane**
Bachelor of Computer Science in Machine Learning — *2024 - Current*

- UQ Computing Society Industry Officer

Curtin College — **Bentley, Perth**
Diploma in Information Technology | Graduated in 2022 with 6.0 GPA — *Feb 2022 – Oct 2022*

Projects

Hugging Face, Google Collab, OpenCV
Internal Dynamics of Vision Transformer Architectures — *July 2025 – current*
- Investigating internal attention dynamics within Vision Transformer (ViT) architectures using pretrained DeiT models.
- Developed a non-invasive pipeline to extract and analyze attention distributions across transformer layers.
- Exploring direct modifications to attention formulations to study effects on representational behavior, in collaboration with Dr. Kasra Khosoussi.

Finetuning Flan-T5 - BioLaySumm — ***PyTorch, Hugging Face Transformers, NVIDIA A100***
- Fine-tuned a pretrained **FLAN-T5-small/base** encoder–decoder model on the **BioLaySumm Subtask 2.1** dataset using full fine-tuning across 4 epochs with mixed-precision training on an NVIDIA A100 GPU (40 GB VRAM).
- Evaluated performance on a held-out test split using **ROUGE-1**, **ROUGE-2**, **ROUGE-L**, and **ROUGE-Lsum** metrics to assess layperson translation quality.
- Conducted error analysis with 5 representative input–output examples, achieving > 70% accuracy across different rouge scores

Bachelor's Capstone Project – TraceForest and StyleCodeGPTSensor — ***Jira, PyTorch, scikit-learn***
- Developed an explainable Random Forest model achieving 94% accuracy for detecting AI-assisted code using stylometric AST features.
- Integrated SHAP-based explainability to identify which stylometric attributes influenced AI vs human predictions.
- Built Python and Java adapters returning label probabilities and feature contributions for automated PR review within secure workflows.
- Upgraded CodeGPTSensor by fine-tuning the pretrained encoder with stylometric features, achieving >95% accuracy.

DAWNBench Challenge — ***PyTorch, CUDA, Google Colab, CIFAR-10***
- Trained a ResNet-style CNN in PyTorch achieving >93% test accuracy on CIFAR-10 within 30 minutes.
- Implemented data augmentation, mixed-precision training, and cosine learning-rate scheduling.
- Optimized GPU utilization for high training efficiency.

OASIS Challenge – UNet Brain MRI Segmentation — ***PyTorch, CUDA, Google Colab***
- Implemented a 2D UNet for multi-class brain MRI segmentation using Cross-Entropy and Dice loss.
- Achieved >90% Dice Similarity Coefficient (DSC) across all labels with early stopping.
- Visualized inference results on Google Colab using NVIDIA A100 GPU (data licensed by UQ).

Persistent Homology ML Pipeline (PH-AML) — ***Julia, Python, Jupyter Notebook***
- Developed a pipeline integrating topological data analysis into supervised learning.
- Extracted persistent homology features (barcodes, landscapes) via Vietoris–Rips filtrations.
- Enhanced classification performance on high-dimensional K-complex datasets.